

MIKHAIL KIM

Python Backend Engineer | Search & AI Systems

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Python backend engineer with 4+ years building cloud, IoT, search and LLM systems. Own services end to end, from API and data design to delivery, monitoring and production reliability.

EXPERIENCE

YADRO | Backend Engineer

Mar 2025 - Present

Speechmate: cloud platform for wearable audio devices.

- Built monitoring for 500+ IoT devices with MQTT, Redis and batched Telegram alerts; cut offline-device detection time from 8 hours to 3 minutes.
- Migrated the cloud core from Sanic to FastAPI with OpenAPI and Pydantic; reduced p95 API latency from 450 to 200 ms and endpoint delivery time from 2 hours to 20 minutes.
- Automated firmware delivery in GitLab CI, cutting commit-to-device time from 4 hours to 50 minutes; standardized Linux networking, reducing incidents from 10 to 1 per month.
- Scaled recording processing from 10 to 80 per hour through batched device uploads and parallel RabbitMQ workers, clearing the backlog and enabling same-day analysis.

NorthCode | Backend Engineer

Apr 2022 - Mar 2025

Procurement catalog search and classification; WAMP developer assistant.

- Built hybrid BM25 and vector search with Reciprocal Rank Fusion for the KTRU procurement catalog; placed the correct code in the top 100 for 19 of 20 queries, up from 9 of 20.
- Improved LLM classification from 6 to 9 correct codes out of 10 over 16 prompt iterations; semantic filtering cut context from 500k to 300k tokens and halved query cost.
- Extracted WAMP's AI engine into a service with a 200-operation JSON-RPC API, EdDSA/JWKS authentication and WebSocket streaming; cut client integration from 5 days to 2 hours.
- Added Prometheus, Grafana and Loki observability; sustained 300 concurrent AI streams with failures below 1% and reduced incident diagnosis from 3 hours to 10 minutes.

TECHNICAL SKILLS

Backend Python, FastAPI, asyncio, SQL, SQLAlchemy, REST APIs, WebSocket, JSON-RPC

Data & AI PostgreSQL, Elasticsearch, ClickHouse, Redis, RabbitMQ, MQTT, OpenAI API

Delivery Docker, Linux, GitLab CI/CD, Nginx, Prometheus, Grafana, Loki, unit testing

EDUCATION & LANGUAGES

Saint Petersburg State Marine Technical University

2022

Software Development, Faculty of Digital Industrial Technologies

Languages English - C1 (advanced) | Russian - native